



COURSE SYLLABUS						
Cours e Code	COMP 007	Course Title	OPERATING SYSTEMS		Course Credit	3
Course Description		This course introduces the concepts, structures, and functions of operating systems, focusing on process, memory, file, and device management. It also provides hands-on experience in using and managing Windows and Linux environments.				
Pre-Requisites		COMP 001	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI
APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
PROMOTION



STARS

★★★★★
Employability
★★★★★
Online Learning
★★★★★
Internationalization

★★★★★
Teaching
★★★★★
Academic
Development
★★★★★
Inclusiveness

★★★★★
Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)



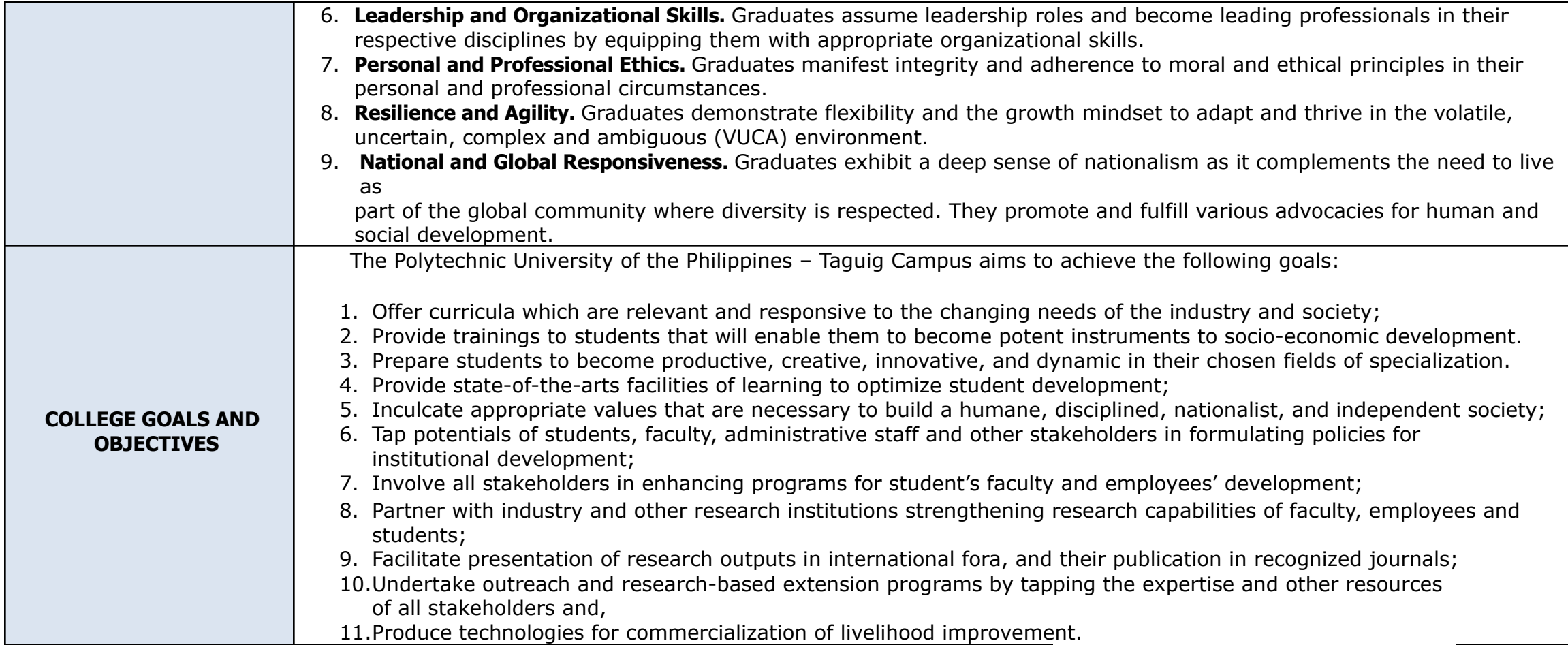
Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.





PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia  The WORLD UNIVERSITY RANKINGS for INNOVATION Rank 21 GENERATIVE AI APPLICATION Rank 39 STUDENT SUPPORT ENGAGEMENT Rank 50 SYMBOL OF PROMOTION  5 Stars Employability Teaching Online Learning Academic Development Inclusiveness Social Responsibility Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)  Times Higher Education Impact Rankings 2024 DEBUTS AT 1501+	5 OLOs <table><tr><td>5</td><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>	5	7	8	9				
	5	7	8	9							
Apply knowledge of computing, science, and mathematics appropriate to th											

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>



	Analyze complex problems, and identify and define the computing requirements appropriate to its solution											
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.											
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.											

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SYMBOL/ PROMOTION

5+ stars

- Employability
- Online Learning
- Internationalization

4+ stars

- Teaching
- Academic Development
- Social Responsibility

3+ stars

- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Impact Rankings 2024

DEBUTS AT 1501+



	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain the fundamental concepts, structures, and functions of operating systems.								
	Identify and differentiate various types of operating systems and their core functions.								
	Analyze process management concepts, including scheduling, synchronization, and deadlock handling.								

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WURI
The World University Rankings for INNOVATION
Rank 21 GENESRATIVE AI APPLICATION
Rank 39 STUDENT SUPPORT ENGAGEMENT
Rank 50 SYMBOL/ PROMOTION

QS STARS
★★★★★
- Employability
- Online Learning
- Internationalization
★★★★★
- Teaching
- Academic Development
- Inclusiveness
★★★★★
- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education Impact Rankings 2024
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	Apply memory management techniques such as paging, segmentation, and virtual memory in problem-solving scenarios.								
	Demonstrate an understanding of file systems, input/output management, and storage organization.								
	Utilize Windows and Linux environments to perform system operations, file management, and process monitoring.								
	Evaluate system security measures, protection mechanisms, and performance considerations in operating systems.								
	Integrate operating system concepts through simulation projects and hands-on activities reflecting real-world applications.								

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Online Learning
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Teaching
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Academic Development
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Internationalization

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OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Understand the basic concepts and importance of Operating Systems.	Definition, Purpose, and Functions of an Operating System; OS Structure.	Lecture and interactive discussion.	Online quiz and asynchronous video lesson on OS roles.	Short quiz / recitation	Online activity submission				
3	Describe the types and classifications of operating systems. Batch, Time-Sharing, Real-Time, Distributed, and Embedded OS.	Batch, Time-Sharing, Real-Time, Distributed, and Embedded OS.	Comparative lecture on OS types.	Group research posting on LI discussion board	 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia  Rank 21 Generative AI Application Rank 39 Student Support Engagement Rank 50 Symbol Promotion  5+ stars - Employability - Teaching - Online Learning - Academic Development - Social Responsibility - Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND) - Inclusiveness - Internationalization  DEBUTS AT 1501+					





	Embedded OS.									
4	Explain the concept of process management.	Process States, PCB, Context Switching, Scheduling.	Illustration of process lifecycle and context switching.	Reflection post about real-life process examples.	Activity / Quiz	Reflection paper				
5	Evaluate CPU scheduling algorithms and their efficiency.	FCFS, SJF, RR, Priority, Multilevel Queue Scheduling.	Whiteboard demo and problem-solving.	Asynchronous coding practice using a simulation tool.	Quiz	Practical coding submission				
6	Understand inter process communication and synchronization.	IPC Methods, Semaphores, Monitors.	Lecture and sample demonstrations.	Watch recorded video and summarize IPC examples.	Oral recitation	Summary submission				
7	Identify and solve deadlock problems.	Deadlock Prevention, Avoidance, Detection, Recovery.	Case analysis and guided problem-solving.	Worksheet on identifying deadlock conditions.	Quiz	Activity worksheet				



8	Demonstrate knowledge of memory management techniques and apply concepts through Windows-based system analysis.	Contiguous Allocation, Paging, Segmentation; Windows OS Overview, File and Process Management, Task Manager, Registry, and System Utilities.	Lecture-demonstration on memory management and Windows OS tools.	Online reading and quiz on memory allocation; hands-on asynchronous exercises using Windows environment.	Seatwork / Midterm Practical Exam	Online quiz / Midterm Written Exam				
9	MIDTERM WRITTEN EXAMINATION									
10	Discuss concepts of virtual memory and paging.	Demand Paging, Page Replacement Algorithms, Thrashing.	Problem-solving discussion.	Simulation of paging algorithm using online tools.	Seatwork	Quiz				
11	Explain file system architecture and management.	File Concepts, Access Methods, Directory Structures.	Discussion and sample exercises.	File system activity using virtual OS.	Activity output	Quiz				
12	Discuss secondary storage and disk management.	Disk Scheduling Algorithms, RAID, Swap Space.	Lecture and examples.	Reading assignment and asynchronous quiz.	Short quiz	Worksheet submission				
13	Explain input/output management and device handling.	I/O Hardware, Device Drivers, Buffering, Spooling.	Lecture and demo of I/O operations.	Watch pre-recorded demo video.	Seatwork	Reflection				
14	Identify and	Security Threats,	Case study presentation.	Research art						





	analyze system protection and security features.	Authentication, Encryption, Access Control.		OS security threats.		activity				
15	Understand file and resource sharing in distributed systems.	Distributed OS Concepts, Client-Server Model.	Lecture and examples.	Group assignment on distributed systems.	Short quiz	Group paper submission				
16	Discuss modern operating systems for mobile and cloud computing.	Android OS, iOS, Cloud OS, Virtualization Concepts.	Lecture and open forum.	Group presentation on mobile OS architectures.	Oral presentation	Uploaded slides / report				
17	Apply OS concepts through hands-on activities and projects; demonstrate understanding and practical use of Linux OS.	Mini Project: Process Scheduling / File Management Simulation; Linux OS Overview; CLI; File Permissions; Shell Commands.	Presentation and peer feedback; Demonstration of Linux environment and terminal commands.	Asynchronous project development; Online quiz and practical tasks using Linux shell simulation.	Project output; Final practical exam.	Video submission/defense; Written exam.				
18	FINAL WRITTEN EXAMINATION ()									





		Resources and Applying Strategies in Teaching Prose Competency 4: Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: GeeksforGeeks. (n.d.). Operating systems. GeeksforGeeks. https://www.geeksforgeeks.org/operating-systems/operating-systems/ OSDev Wiki. (n.d.). OSDev – The OS development wiki. https://wiki.osdev.org Operating-System.org. (n.d.). Operating-System.org. https://www.operating-system.org/ The Linux Documentation Project. (n.d.). The Linux Documentation Project (LDP). https://tldp.org/ OSNews. (n.d.). OSNews. https://www.osnews.com/ WORLD UNIVERSITY RANKINGS 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia The WORLD UNIVERSITY RANKINGS for INNOVATION Rank 21 GENERATIVE AI APPLICATION Rank 39 STUDENT SUPPORT ENGAGEMENT Rank 50 SYMBOL/ PROMOTION STARS 5 5 5 • Employability • Online Learning • Internationalization • Teaching • Academic Development • Inclusiveness • Social Responsibility • Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND) Times Higher Education Impact Rankings 2024 DEBUTS AT 1501+										





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STARS



- Employability
- Online Learning
- Internationalization



- Teaching
- Specialist Courses
- Inclusiveness



- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BNDT)



WU
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Rank 21
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Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL/ PROMOTION



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TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)





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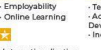
Rank 50
SYMBOL/
PROMOTION



5+ stars

5+ stars

5+ stars



Employability

Teaching

Social Responsibility

Online Learning

Academic Development

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Internationalization

Inclusiveness



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class p
coursed through university-issued (i.e., official) email accounts. Stu

Intellectual Property and Copyright.
Original materials created for this online class is covered by ur
(third-party) instructional materials (e.g., YouTube videos) might l
while others (e.g., some assigned readings) might be subject to f



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	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	30%	Final Examinations (Teaching Demonstration)	30%
	100%		100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.		
COURSE REQUIREMENT/S	GRADING SYSTEM		
Class participation and attendance in all sessions. Quizzes and seatwork on key operating system topics. Laboratory and hands-on activities using Windows and Linux. Individual or group assignments related to OS concepts. Midterm examination focused on Windows OS. Final examination focused on Linux OS. Final project applying OS principles with presentation or defense.	Final Grade = (Midterm Grade + Final Term Grade) / 2		





CLASS INFORMATION	FACULTY INFORMATION
Course, Year and Section: BSIT 2-1	Name of Faculty: FRANCO, FRANCIS FRADES
Day and Time: THURSDAY AND 7:30AM – 12:30PM	Consultation Time:
Room: DOST LAB	Office Tel. No./ Mobile Phone No.
School Year: 2025 – 2026 Semester: 1 ST SEMESTER	E-Mail Address:

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses



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